

What does your pressure space miss?

Compare a prescribed log-volume field, its pressure-space projection, and the component that the space cannot represent.

A representation lesson, at a fixed state.

This is a finite weighted example. There is no body deformation, equilibrium solve, or incompressibility simulation. Refinement changes the space, while the prescribed field stays the same.

Full sample energy

8

$$E_{full} = K/2 \sum w_i g_i^2$$

Condensed energy

0.5

$$E_{cond} = K/2 \sum w_i (Pg)_i^2$$

Unresolved energy gap

7.5

$$E_{full} - E_{cond} = K/2 \sum w_i r_i^2$$

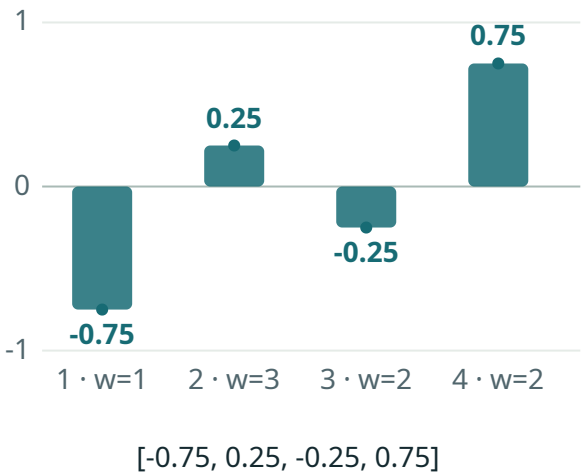
The field, its projection, and the remainder

Discrete samples · one shared vertical scale · no material geometry

Prescribed field

g

Authored log-volume samples



Represented part

Pg

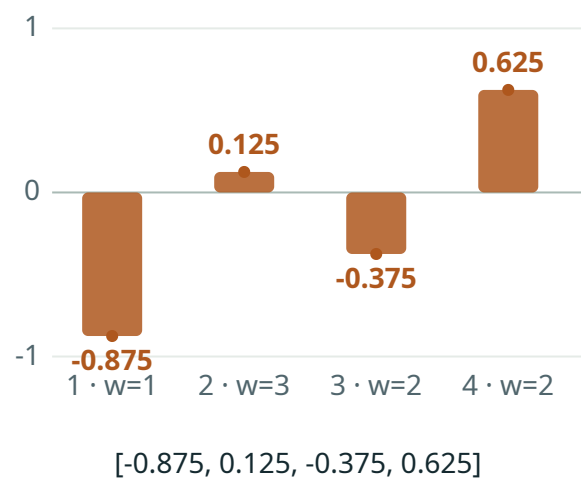
One weighted mean, repeated



Unresolved part

r

$r = g - Pg$



Actual values used in this calculation

Sample	w_i	g_i	$(Pg)_i$	r_i	$p^*_i = K(Pg)_i$
1	1	-0.75	0.125	-0.875	1
2	3	0.25	0.125	0.125	1
3	2	-0.25	0.125	-0.375	1
4	2	0.75	0.125	0.625	1

p^* is the optimal mixed variable in the chosen sample space, not a measured anatomical pressure.

Full: $4 \times 2 = 8$
Condensed: $4 \times 0.125 = 0.5$
Residual: $4 \times 1.875 = 7.5$
Mixed objective at p^* : 0.5

Readouts round to five decimal places; the identity diagnostic uses unrounded values.

Weighted residual sums by group: [0]. Browser identity difference: 0.

What refinement recovers

Every row uses the same current g , weights, and K .

Pressure values	Econd	Gap
● 1 shared	0.5	7.5
● 2 groups	1	7
● 4 samples	8	0

Refining from one shared value to two groups recovers the pair means. Four sample values recover this entire prescribed field, with zero gap.

Energy ordering is at a fixed state. It does not establish stability or compare solutions re-equilibrated in different spaces.

THE PROJECTION IS EXPLICIT

$$(Pg)_i = \sum_{j \in G_i} w_j g_j / \sum_{j \in G_i} w_j$$

G_i is the selected group containing sample i . Q consists exactly of vectors constant on each group. Positive weights make each group denominator positive.

WHY THIS IS ORTHOGONAL

$$\sum_{j \in G} w_j (g_j - (Pg)_j) = 0$$

The unresolved part has zero weighted sum in each group. It is therefore weighted-orthogonal to every group-constant vector in Q . Ordinary unweighted averages would give a different projector.

Exact algebra and browser arithmetic are separate.

The 27 Lean statements prove weighted projection, unique maximizing pressure vectors, condensed energy, the nonnegative gap, and nested-space monotonicity under exact assumptions. This lab checks its numerical implementation separately.

[Read the standalone proof guide and exact reference](#) · [Lean source](#)

Interpretation and limits

g is prescribed, not solved. Interpreting its entries as $\log J$ requires positive J and justified sampling; no continuum or incompressibility result follows. K is one positive constant. The Lean projector is assumed exact. Pressure-vector uniqueness does not certify arbitrary basis-coefficient uniqueness, which needs injective evaluation. No nonlinear equilibrium, inf-sup stability, derivative/Hessian gap, anatomical validity, or floating-point refinement proof is claimed.